Orbital mechanics and corpus linguistics rule out the Star of Bethlehem

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Abstract

The Star of Bethlehem (Matthew 2:9) depicts a celestial body guiding travellers southward for two hours, then halting above a specific house. Rather than identifying the best candidate event, we derive kinematic constraints from the text and show the feasible set of orbital solutions is empty. A heliocentric survey of 529 million orbital configurations—spanning elliptic, parabolic, and hyperbolic orbits across all inclinations, eccentricities, perihelion distances, and orientations, tested at every major proposed epoch from 12 BCE to 1 BCE—finds zero satisfying sustained guidance motion ($>2°$/h), punctual stopping ($<2°$/h), and the required azimuth corridor simultaneously. Monte Carlo refinement (2.25 million draws near the best-performing orbits), geocentric surveys of 51,200 Earth-orbiting configurations, and a close-flyby analysis of 10 million comet orbits confirm the null result. The impossibility is kinematic: no heliocentric body can maintain the required southward angular motion and subsequently halt above a fixed point. An independent corpus-linguistic analysis of Matthew's four motion terms (προάγω, ἵστημι, ἐπάνω, ἔρχομαι) finds all four absent from astronomical discourse across 25 Greek texts ($\approx \mathbf{622{,}000}$ words; Fisher combined $\boldsymbol{p \approx 6.3 \times 10^{-3}}$), while all four are characteristic of guiding-star legendary narratives (joint Bayes factor $\boldsymbol{\geq 10^{7}}$ under maximally conservative assumptions; baseline $\boldsymbol{\approx 5.3 \times 10^{8}}$). Two independent lines of evidence converge: Matthew 2:9 is a legendary narrative, not a record of an astronomical event.

Keywords: Star of Bethlehem, orbital mechanics, corpus linguistics, Matthew 2:9, archaeoastronomy, ancient Greek astronomy

Testing whether a historical narrative records a real celestial event is ordinarily a search problem: which known sky object best fits the account? Here we show that such questions can instead be resolved by *demonstration*: by exhaustively surveying orbital parameter space and demonstrating that the feasible set is empty. We apply this paradigm to the most studied ancient astronomical claim in the Western record and show that two independent lines of evidence—orbital mechanics and corpus linguistics—converge on the same verdict.

The Star of Bethlehem debate was placed on a renewed scientific footing when Hughes proposed in these pages that the triple Jupiter–Saturn conjunction of 7 BCE was the most probable candidate[1, 2], a paper that has accumulated more than 400 citations and sustained the scientific conversation for five decades. Other proposals include Halley's Comet[3], a nova[4, 5], the retrograde motion of Jupiter in Aries[6], and a 5 BCE comet whose possible orbital elements have most recently been reconstructed from East Asian records[7, 8]. Systematic reviews show no single candidate satisfies all criteria simultaneously[9], and the majority position in New Testament criticism holds that Matthew's infancy narrative is legendary rather than historical[10], though active discussion remains within biblical studies (cf. [11]). What has been lacking is not a better candidate but a general demonstration that no orbital solution of any class can satisfy the text's kinematic requirements, and a quantitative test of whether Matthew's vocabulary belongs to the astronomical register at all.

The critical constraints come from Matthew 2:9: ὁ ἀστὴρ οὗ εἶδον ἐν τῇ ἀνατολῇ προῆγεν αὐτούς, ἕως ἐλθὼν ἐστάθη ἐπάνω οὗ ἦν τὸ παιδίον ("the Star which they had seen at its rising was going before them, until, having arrived, it stood over where the child was"). The verb προῆγεν (imperfect active of προάγω) denotes sustained forward motion ahead of the travellers throughout their journey. The form ἐστάθη (aorist passive of ἵστημι), bounded by the temporal particle ἕως ("until"), denotes a completed, punctual coming-to-rest—a change of state, not a prior condition[9]. The phrase ἐπάνω οὗ ("over where") further specifies that the stopping occurred directly above a particular dwelling[9, 12, 13]. The two verbs define two physically distinct phases of apparent motion: a *guidance phase* in which the Star moves ahead of travellers walking from Jerusalem to Bethlehem, and a *stopping phase* in which it becomes motionless above a specific house.

Three converging lines of evidence are presented in the order that builds the argument most naturally. First, Matthew's own usage of the four motion words across his other 118 Gospel occurrences—zero in any astronomical sense—yields an internal Bayes factor of $\approx 88{,}000$ against the astronomical reading. Second, orbital mechanics closes the loose-language escape: the impossibility is fundamentally geometric—objects in the road corridor appear quasi-stationary (satisfying the stopping criterion but not guidance), while objects with sufficient southward motion lie outside it; no Keplerian body can satisfy both simultaneously. Third, external corpus-linguistic analysis of 25 Greek astronomical texts ($\approx 622{,}000$ words) places Matthew's vocabulary in the legendary guiding-star tradition (BF $\approx 6{,}000$). The linguistic datasets are disjoint; their joint product exceeds 10^7 under maximally conservative assumptions (baseline $\approx 5.3 \times 10^8$).

1 Results

1.1 Matthew's own vocabulary excludes the astronomical reading

A necessary condition for the astronomical reading of Matthew 2:9 is that the four motion words can carry astronomical senses. The author's own usage elsewhere in his Gospel provides direct evidence against this: across all 118 other occurrences of προάγω, ἵστημι, ἐπάνω, and ἔρχομαι in Matthew, zero appear in any sense compatible with an astronomical body (Methods; full passage data in Supplementary Information).

Προάγω appears five more times (14:22; 21:9×2; 26:32; 28:7), always transitive with a human direct object—identical syntax to προῆγεν αὐτούς in Matt 2:9; the intransitive positional-antecedence sense the astronomical reading requires is unattested in Matthew, nor does Matthew use other terms for "leading" that have an astronomical register (e.g., the intransitive προηγέομαι). Ἵστημι appears 20 more times with persons or objects at named physical locations; the mathematical stationary point is never its referent, nor does Matthew use a verb more commonly found in astronomical texts (e.g., στηρίζω). Ἐπάνω appears seven more times denoting physical contact or immediate proximity; at Matt 28:2 an angel sits ἐπάνω the rolled-away stone—a heavenly being in point-contact with a terrestrial object, the same register as Matt 2:9. Ὑπέρ would be the expected Greek preposition for an astronomical body. On its own, read literally, ἐπάνω disqualifies the astronomical reading: a celestial body cannot be "over" a specific house in the sense of physical contact or immediate proximity. Nonetheless, I will conservatively only consider word counts in Matthew's own idiolect, not the semantic content of the words themselves, in the Bayes factor calculation below. Ἔρχομαι appears 86 more times with persons or agents arriving at terrestrial locations; never with a celestial body reaching a zodiacal position or celestial coordinate. Another verb could be used for "arriving" in astronomical texts (e.g., καταντάω), but Matthew never uses it.

Using Laplace smoothing, the Bayes factor for feature i from Matthew's own idiolect is $\mathrm{BF}_i = n_i + 1$ (non-astronomical uses n_i, zero astronomical uses). The joint within-Matthew Bayes factor is:

$$\mathrm{BF}_{\mathrm{Matthew}} = \prod_i (n_i + 1) = 6 \times 21 \times 8 \times 87 = 87{,}696. \tag{1}$$

Matthew's own vocabulary disfavors the astronomical reading at ≈88,000 to 1.

The key epistemic point is what $\mathrm{BF}_{\mathrm{Matthew}}$ measures: not that Matthew never discusses astronomy elsewhere—he rarely has occasion to—but that when describing the Star's motion he consistently reaches for words the external corpus identifies as belonging to the non-astronomical register. This reading is not a modern analytical artifact: no ancient or medieval commentator interpreted the Star's motion in Matt 2:9 as a natural astronomical phenomenon[16]; every pre-modern interpreter understood the Star as not astronomical or astrological, but miraculous. The following sections test the remaining escape route: whether colloquial or imprecise language could nonetheless describe a real celestial event.

1.2 Geometric and temporal constraints from Matthew 2:9

The Magi traveled from Jerusalem (Old City center: $31°46'$ N, $35°14'$ E) to Bethlehem (Church of the Nativity: $31°42'$ N, $35°12'$ E), approximately 8 km along the ancient ridge road to the south-southwest. The straight-line bearing is 203°; because the road deviates around the central ridge, the actual azimuth of travel sweeps from approximately 190° to 210° along the route. I therefore adopt a road-corridor guidance criterion: the Star's azimuth must remain within $[190°, 210°] \pm 5°$ throughout the 2-hour journey.

The criterion for προῆγεν ("was going before") is not merely positional: a stationary object at the right azimuth cannot "go before" (let alone guide) travellers[9]. I therefore require the Star to exhibit perceptible southward motion throughout the guidance phase. I adopt an angular-velocity floor of $\omega_{\text{thresh}} = 2°\,\text{h}^{-1}$ ($4\times$ the Moon's motion relative to background stars, $\approx 0.5°\,\text{h}^{-1}$, which Babylonian astronomers monitored routinely). This threshold is deliberately generous: naked-eye resolution is approximately $1'$, so motion is detectable over a 30-minute observation at any rate above $\approx 0.03°\,\text{h}^{-1}$, thirty-fold below our floor. Choosing any stricter threshold only strengthens the impossibility conclusion. The *guidance criterion* is therefore: (i) azimuth within the road corridor for the full journey, and (ii) apparent angular velocity $> 2°\,\text{h}^{-1}$ throughout.

Walking pace on hilly terrain yields a journey time of approximately 2 hours. I locate this journey in the pre-dawn hours of a date in 5 BCE (Julian Day 1 719 755.9, $\approx$ 08:00–10:00 local solar time), consistent with the most recently proposed candidate[8]. Following arrival at Bethlehem, the Star must appear stationary over the specific house (ἐστάθη ἐπάνω, "stood over"). The *stopping criterion* is maximum apparent angular velocity below $2°\,\text{h}^{-1}$ during the 10:00–12:00 window immediately following the journey—the same threshold ω_{thresh}, applied as an upper rather than lower bound. A sensitivity sweep across thresholds $\omega_{\text{thresh}} \in \{0.25, 0.5, 1, 2, 5, 10\}°\,\text{h}^{-1}$ yields zero satisfying configurations at every value tested (Supplementary Information, Table S1).

1.3 Kinematic demonstration of incompatibility

The argument in this subsection is a mathematical demonstration, not a numerical result. The conceptual core is geometric: the south-southwest bearing of the Jerusalem–Bethlehem route means that any Keplerian body within the road corridor appears quasi-stationary throughout (satisfying the stopping criterion but not guidance), while any body with sufficient southward apparent motion lies outside the corridor—a mutual exclusion. The analysis below makes this precise, establishing from the geometry of the bearing and Earth's rotation rate alone that the required ICRS angular velocities during guidance and stopping are unachievable for any heliocentric body, and that the transition between them is additionally forbidden by gravitational dynamics. The numerical surveys in the following subsections are corroborative.

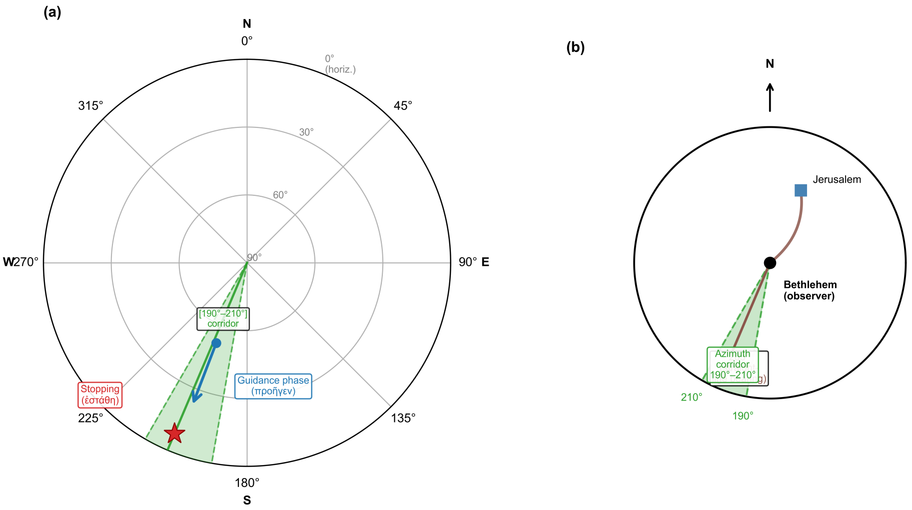

Fig. 1 Observational constraints from Matthew 2:9. a, Polar sky chart showing the required apparent motion: during the 2-hour guidance phase (προῆγεν, "was going before"), the Star must move visibly southward with azimuth within the road corridor [190°, 210°] toward Bethlehem; it then appears stationary over a specific house (star symbol, ἐστάθη, "stood over"). Altitude is not constrained during stopping (any altitude above the horizon is acceptable)—scholarly interpretations of "stood over" range from just above the horizon to zenith. **b**, The road corridor [190°, 210°] in the local sky from Bethlehem. The straight-line bearing is 203°; the road deviates around the central ridge, sweeping between 190° and 210°.

Converting the two observational phases into ICRS velocity requirements via pre-computed East–North–Up (ENU) rotation matrices (Methods), the argument proceeds as follows.

*Stopping phase (*10:15–12:00*).* An object motionless in the local horizontal frame ("stood over") must, because the local AltAz frame co-rotates with Earth, sweep through ICRS at the sidereal rate. At a representative stopping altitude of 45° (az = 200°, Bethlehem latitude 31.7°N), this is $\omega_{\mathrm{stop}} = 14.76°\ \mathrm{h}^{-1}$ (Table 1). Such a criterion already excludes fixed stars, novae, and planets, which all move at approximately the sidereal rate.

*Guidance phase (*08:00–10:00*).* An object visibly "going before" travellers at the minimum rate $2°\ \mathrm{h}^{-1}$ in local AltAz along azimuth 200° yields, by vectorial addition of the sidereal rate, $\omega_{\mathrm{guide}} = 15.55°\ \mathrm{h}^{-1}$ in ICRS.

Primary impossibility (literal interpretation). Both ICRS values ($\approx 15°\ \mathrm{h}^{-1}$) are unachievable for standard heliocentric bodies. Since $\omega_{\mathrm{ICRS}} = V_\perp/d$ with $V_\perp \propto d$, for any naked-eye heliocentric candidate with $d \gtrsim 0.1\,\mathrm{AU}$, the stopping state requires $V_\perp \gtrsim 1{,}070\,\mathrm{km\,s^{-1}}$ (e.g. $\approx 5{,}560\,\mathrm{km\,s^{-1}}$ at Mars opposition; $\approx 44{,}900\,\mathrm{km\,s^{-1}}$ at Jupiter)—far above any known solar-system speed. The very close flyby regime ($d \sim 0.003\,\mathrm{AU}$) reduces the requirement to $V_\perp \sim 30\,\mathrm{km\,s^{-1}}$ (achievable in principle), but apparent motion there is overwhelmingly diurnal—indistinguishable from any fixed star—and Matney's proposed comet occupies a fully sunlit sky throughout the relevant window. A dedicated nighttime close-flyby survey of 10 million Monte Carlo orbits (Supplementary § S5.5) confirms that zero of 66 genuine close-flyby orbits satisfy the guidance → stopping criterion; a numerical optimizer (100 restarts) finds no orbit achieves more than 0.58 h of contiguous guidance duration (requirement: 1.0 h). Full derivation and distance table in Supplementary § S1.9.

Secondary constraint (astronomical-vocabulary interpretation). Even under the more permissive reading that the verbs describe motion relative to the stellar background, the transition between the two states is forbidden. The guidance and sidereal-tracking trajectories differ in direction in ICRS, requiring an impulsive $\Delta v \approx 1.5\,\mathrm{km\,s^{-1}}$ at $d = 0.0026\,\mathrm{AU}$. Under Earth gravity ($a_\oplus = 2.6 \times 10^{-6}\,\mathrm{km\,s^{-2}}$) this requires $\approx 158\,\mathrm{h}$; the deceleration time scales as r^3, making the problem strictly worse at any greater distance. The corpus-linguistic analysis below independently closes this interpretive retreat by showing that the verbs belong to the narrative, not astronomical, register.

Table 1 Kinematic parameters for the guidance and stopping phases at three representative stopping altitudes (azimuth 200°, Bethlehem). ω_{guide}: ICRS angular velocity of a star rising at the minimum detectable rate ($2°\,\mathrm{h}^{-1}$ in local AltAz) along azimuth 200°. ω_{stop}: ICRS angular velocity of a star stationary in local AltAz (sidereal tracking). Δv at $r = 0.0026\,\mathrm{AU}$ (Matney's closest approach): minimum transverse velocity change required for the phase transition.

Alt. (°)	ω_{guide} ($°\,\mathrm{h}^{-1}$)	ω_{stop} ($°\,\mathrm{h}^{-1}$)	$\Delta\omega$ ($°\,\mathrm{h}^{-1}$)	$\omega_{\mathrm{stop}}/\omega_{\mathrm{thresh}}$	Δv ($\mathrm{km\,s^{-1}}$)
20	13.44	12.34	1.10	6.2×	2.07
45	15.55	14.76	0.79	7.4×	1.50
60	15.68	15.02	0.66	7.5×	1.24

1.4 Exhaustive survey of heliocentric orbits

The null result is driven primarily by the geometric incompatibility between sustained southward motion and corridor confinement; the stopping criterion then eliminates the small residual near-matches that pass azimuth screening. I conducted an exhaustive heliocentric survey with a fine angular grid. The longitude of ascending node Ω and argument of perihelion ω were sampled at 5° intervals ($72 \times 72 = 5{,}184$ orientations per outer combination) across eccentricities $e \in \{0, 0.1, 0.2, 0.3, 0.5, 0.7, 0.8, 0.9, 0.95, 0.99, 1.0, 1.1, 1.5, 2.0, 5.0\}$ (elliptic through hyperbolic, spanning the eccentricity range of the known interstellar objects 1I/'Oumuamua ($e \approx 1.20$)[14] and 2I/Borisov ($e \approx 3.36$)[15]), perihelion distances $q \in \{0.02, 0.03, 0.04, 0.05, 0.07, 0.10, 0.15, 0.20, 0.25, 0.30, 0.35, 0.40, 0.50, 0.60, 0.70, 0.80, 0.90, 1.00\}$ AU (18 values, reaching Earth's orbital distance), inclinations 0° to 180° in 5° steps with additional fine sampling near 0° (42 values total; Supplementary Information §S1.7), and perihelion epoch offsets $\Delta T \in \{-60, -40, -20, -10, -5, +5, +10, +20, +60\}$ days (9 values), yielding $18 \times 15 \times 42 \times 9 = 102{,}060$ outer combinations and $102{,}060 \times 5{,}184 = 529{,}079{,}040$ total configurations sampled. Of 202,372,076 visible configurations (object above 5°

altitude throughout the guidance window), zero simultaneously satisfy all three criteria: road-corridor azimuth, guidance motion $> 2^\circ\,\mathrm{h}^{-1}$, and stopping motion $< 2^\circ\,\mathrm{h}^{-1}$ (Table **Extended Data Table 1**). The best azimuth corridor score is 0.015° inside the corridor (orbit $e = 1.1$, $q = 0.03$ AU, $i = 10^\circ$, identified by the complementary fine-grid orbit inversion of Section 1.6), yet that configuration's angular velocity is $0.23^\circ\,\mathrm{h}^{-1}$—an order of magnitude below the required guidance motion. Configurations that come closest to the azimuth criterion prove nearly stationary: they satisfy the stopping criterion but have angular velocities far below the $2^\circ\,\mathrm{h}^{-1}$ guidance-motion threshold, because a stationary object cannot προάγειν ("go before") anyone.

To verify this null result is not a grid-spacing artefact, I performed two Monte Carlo refinement sweeps (Supplementary Information §S1.7 and Fig. S1). Sweep 1 (1.5 million configurations near the best-fit grid orbit) found zero of 601,150 visible configurations satisfying all criteria; the closest azimuth match had guidance velocity of $0.006^\circ\,\mathrm{h}^{-1}$—a factor of $\sim$330 below threshold. Sweep 2 (750,000 configurations near Matney's proposed 5 BCE comet) found zero of 294,672 visible configurations, with the best-azimuth candidate lying outside the road corridor entirely. No fine-tuning yields a viable solution: the incompatibility is structural.

1.5 Survey of Earth-orbiting objects

For geocentric objects the primary impossibility of Section 1.3 does not apply: an Earth-orbiting body *can* in principle exhibit perceptible motion and then appear to slow, because its apparent motion is governed by its orbital velocity relative to the ground observer rather than by the sidereal rate. The geocentric case therefore requires a separate analysis in ground-frame AltAz coordinates.

The Moon—the only known natural geocentric body in antiquity—reaches $\omega_{\max} = 0.61^\circ\,\mathrm{h}^{-1}$ at perigee, falling short of the $2^\circ\,\mathrm{h}^{-1}$ guidance threshold by a factor of three. A comprehensive survey of 28,800 hypothetical geocentric orbit configurations (the physically admissible subset—perigee altitude $> 100\,\mathrm{km}$—of 51,200 total grid configurations; semi-major axes from 7,000 km to 500,000 km, eccentricities 0–0.99, all inclinations and orientations; Supplementary § S1.10) finds zero configurations satisfying all criteria with a realistic stopping threshold $\omega_{\mathrm{stop}} < 0.20^\circ\,\mathrm{h}^{-1}$. The only configurations approaching even the loose threshold ($\omega_{\mathrm{stop}} < 0.50^\circ\,\mathrm{h}^{-1}$) are equatorial orbits near geosynchronous distance exhibiting the meridian-transit azimuth artifact: as any equatorial object crosses the meridian, its azimuth rate passes through zero briefly while altitude continues to change—the same geometric effect described for Matney's comet, not a physical deceleration. All 28,800 physically admissible geocentric configurations have the object above 5° altitude during at least one evaluated interval, so all contribute to the visible count. Combined with the heliocentric survey, 202,400,876 visible configurations spanning all orbit types yield zero solutions (Extended Data Table 1; Figs. 2 and 3).

Figure 2 demonstrates the two-criterion near-impossibility: satisfying the road-corridor direction *and* the guidance-motion requirement simultaneously is already unachievable. Figure 3 shows that adding the stopping criterion completes the impossibility. The characteristic slope-1 diagonal in that figure is a rigorous consequence of Kepler's second law: apparent angular velocity $\omega_{\mathrm{app}} = V_\perp/d$ changes by at most $\lesssim 4\%$

over any four-hour window for bodies with $d \gtrsim 0.5\,\mathrm{AU}$ (Supplementary § S1.8), so the sole mechanism that could break this conservation—a retrograde stationary point—is geometrically excluded by the azimuth constraint.

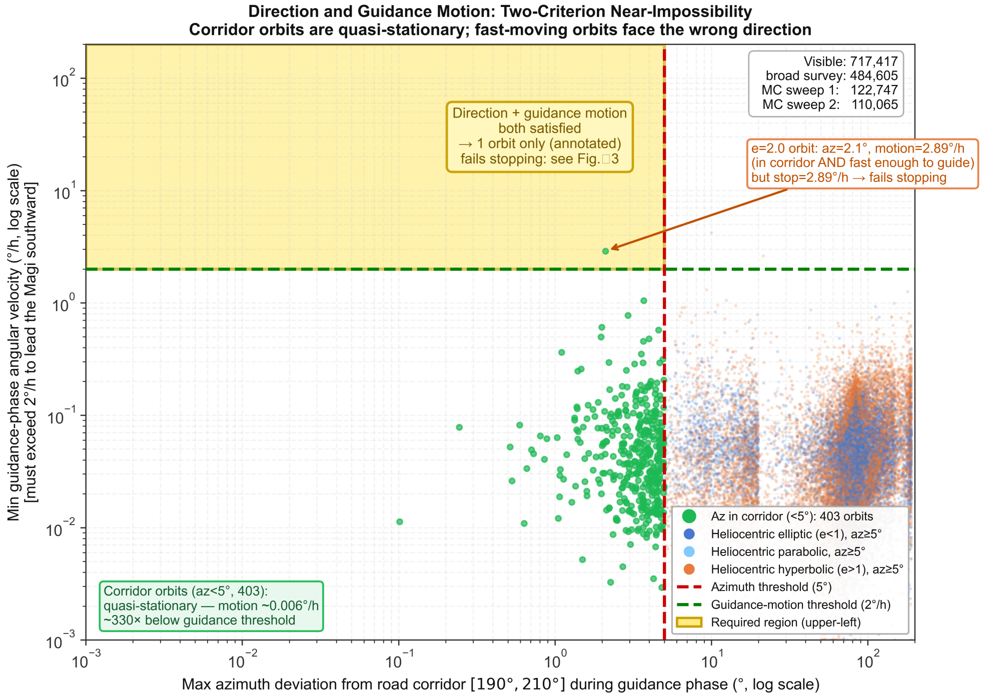

Fig. 2 Direction and guidance motion: two-criterion near-impossibility. Each point is one visible orbital configuration from the full heliocentric parameter space (broad survey) and two Monte Carlo refinement sweeps. The horizontal axis (log scale) shows the maximum azimuth deviation of the object from the road corridor $[190°, 210°]$ during the guidance phase (0 = perfectly in corridor); the vertical axis (log scale) shows the minimum angular velocity during that same phase (must exceed $2°\,\mathrm{h}^{-1}$ to προάγειν). A configuration satisfying both direction *and* guidance motion simultaneously would fall in the gold-shaded upper-left region (azimuth deviation $< 5°$ *and* motion $> 2°\,\mathrm{h}^{-1}$). That region is essentially empty: green points (corridor orbits, azimuth deviation $< 5°$) cluster along the bottom-left, quasi-stationary throughout with guidance speeds $\sim 0.006°\,\mathrm{h}^{-1}$ (factor ~ 330 below threshold); fast-moving orbits (upper right) are far outside the road corridor. The single annotated outlier ($e = 2.0$, $\Delta\mathrm{az} = 2.1°$, motion $= 2.89°\,\mathrm{h}^{-1}$) does enter the gold region but fails the third (stopping) criterion, as shown explicitly in Fig. 3.

1.6 Orbit inversion confirms incompatibility

As a complementary test, I searched for the orbit that best fits the guidance-phase sky positions, then propagated it into the stopping window. Using a fine grid ($5°$ spacing in Ω and ω) over the full eccentricity range ($e = 0$–5.0), the best-fitting guidance orbit achieves an azimuth corridor score of $0.015°$—within the corridor—yet fails the guidance-motion criterion by an order of magnitude ($0.23°\,\mathrm{h}^{-1}$ versus the

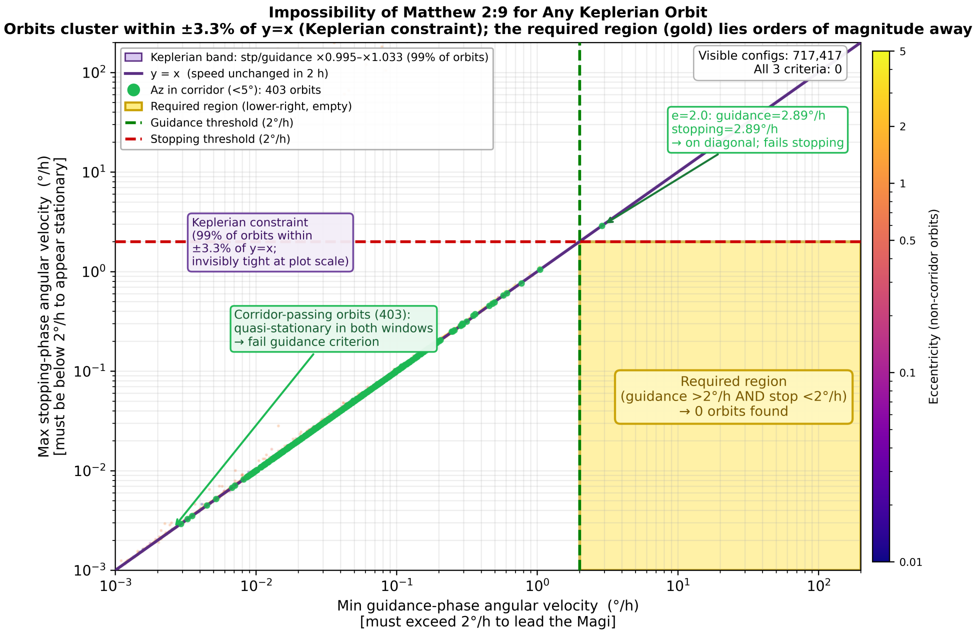

Fig. 3 **Orbital impossibility of Matthew 2:9.** Each point represents one visible orbital configuration sampled from the full heliocentric parameter space (broad survey) and two Monte Carlo refinement sweeps; non-corridor points are coloured by eccentricity (plasma scale). The horizontal axis (log scale) shows the *minimum* angular velocity during the guidance phase (08:00–10:00 local time; must exceed $2^\circ\ \mathrm{h}^{-1}$ to lead the Magi southward). The vertical axis (log scale) shows the *maximum* angular velocity during the subsequent stopping phase (10:15–12:00; must fall *below* $2^\circ\ \mathrm{h}^{-1}$ to appear stationary over Bethlehem). A configuration satisfying προάγειν and ἐστάθη simultaneously would occupy the gold-shaded *lower-right* region; that region contains zero orbits across every eccentricity class. The purple diagonal band is the *Keplerian constraint*: the 1st–99th percentile of the empirical stopping-to-guidance speed ratio, spanning only ×0.995–×1.033 around $y = x$. Over any two-hour window the apparent angular velocity $\omega_{\mathrm{app}} = V_\perp/d$ changes by at most $\lesssim 4\%$ for heliocentric bodies (Supplementary § S1.8); all points therefore cluster invisibly close to the diagonal. The required region lies orders of magnitude from this band, making the exclusion structural rather than a consequence of sparse sampling. Green points (azimuth deviation $< 5^\circ$, in-corridor) cluster near the origin: quasi-stationary throughout both phases, confirming the anti-correlation visible in Fig. 2. The single annotated outlier ($e = 2.0$, $\Delta\mathrm{az} = 2.1^\circ$) reaches $2.89^\circ\ \mathrm{h}^{-1}$ during guidance but retains the same speed during stopping, placing it on the diagonal and well above the stopping threshold. Supplementary Fig. S1 shows each MC sweep separately.

required $2^\circ\ \mathrm{h}^{-1}$). When this orbit is propagated into the stopping window, the object rapidly sets below the horizon—the antithesis of the text's requirement (Extended Data Fig. 1).

1.7 Corpus-linguistic analysis: Matthew 2:9 vocabulary is incompatible with Greek astronomical and astrological discourse

The physical impossibility established above is independently corroborated by a statistical analysis of ancient Greek astronomical vocabulary. If Matthew 2:9 described a real astronomical phenomenon, its language should conform to the technical register shared by Greek astronomical and astrological texts. I assembled a corpus of 25 texts—mathematical astronomy, astrological application, and didactic popularizations—comprising approximately 622,000 words, spanning the fourth century BCE through the fourth century CE (Supplementary Table S7; Methods).

I identified four genre-register features of Matthew 2:9 that any astronomical interpretation must implicitly assume—testing not merely lexical choice but whether the narrative events they encode appear anywhere in astronomical writing:

(A) προάγω (*proagō*): a star leading human travellers toward a destination.
(B) ἵστημι (*histēmi*): rather than the technical στηρίζω: a celestial body coming to rest, not undergoing a planetary stationary point.
(C) ἐπάνω (*epanō*): rather than ὑπὲρ: a celestial body localized above a specific terrestrial address.
(D) ἔρχομαι (*erchomai*): a celestial body "arriving" at a specific location on Earth.

All four Matthew-type usages score zero in the 622,000-word corpus (Table 2; passage-level citations in Supplementary Table S8). Feature A: all 9 corpus occurrences of προάγω carry positional or temporal senses ("preceding in longitude"), never narrative guidance ($p = 0.100$). Feature C: all 29 occurrences of ἐπάνω carry geometric or zodiacal senses, never localizing a body above a named place ($p = 0.033$). Features B and D are likewise absent ($p = 0.056$, 0.125; a thematic robustness check yields the same result at every feature, Supplementary Information, Table S9). The joint Fisher combined statistic is $\chi^2(8) = 21.4$, $p \approx 6.3 \times 10^{-3}$ (Fig. 4); Gaussian copula correction (2×10^6 Monte Carlo draws) maintains significance up to inter-feature correlation $\rho \approx 0.77$, and every drop-one-feature subset remains jointly significant ($p < 0.05$).

Table 2 Corpus linguistics: Fisher's exact test of Matthew 2:9 vocabulary against Greek astronomical and astrological texts.

Feature	Matthew usage	Corpus count	Matthew	p (Fisher)
(A) προάγω	Guidance of travellers	0 / 9	1	0.100
(B) στηρίζω†	Celestial body halts at location	0 / 17	1	0.056
(C) ἐπάνω	Above specific earthly point	0 / 29	1	0.033
(D) ἔρχομαι	Spatial arrival of Star	0 / 7	1	0.125
Joint (A–D)	—	—	—	$\boldsymbol{\chi^2(8) = 21.4}$, $\mathbf{p \approx 6.3 \times 10^{-3}}$

Corpus: 25 texts, ≈ 622,000 words, 4th c. BCE–4th c. CE (see Supplementary Table S7). TLG-verified lemma counts, 2026-05-10 (MIT IRIS proxy); see Supplementary Table S8 for passage-level citations. Fisher's exact test, one-sided: $p = 1/(N+1)$, where N = corpus negatives. "Corpus count" shows Matthew-type positive / non-Matthew-type negative hits. All four Matthew-type usages score zero in the corpus. † στηρίζω serves as proxy for ἵστημι (B) because TLG's morphological expander cannot resolve $\mu\iota$-verb inflections; additional ἵστημι hits would only add B-negatives, making the test more stringent. Joint row reports Fisher's combined $\chi^2 = -2\sum \ln p_i$ with df $= 2k$; corrected for inter-feature correlation by Gaussian copula: $\rho = 0.5$ yields $p = 0.028$; significance maintained up to $\rho \approx 0.77$ (Supplementary Information, Section 3).

The null result is uniform across genre: even astrological texts (Ptolemy's *Tetrabiblos*, Vettius Valens, Hephaestion), the most likely to adopt narrative vocabulary, use φέρεσθαι/κινεῖσθαι and ὑπὲρ throughout; Matthew's guidance vocabulary is absent from mathematical and astrological texts alike (Supplementary Information).

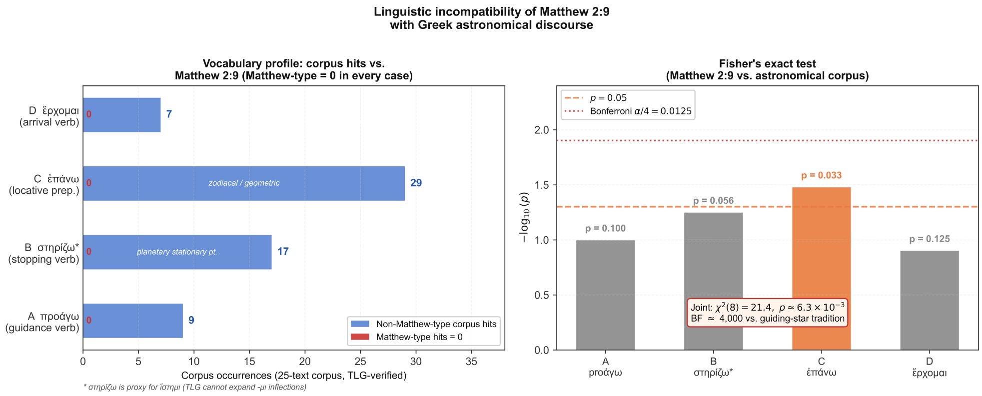

Fig. 4 Linguistic incompatibility of Matthew 2:9 with Greek astronomical discourse. TLG-verified lemma counts (2026-05-10). **Left**, Corpus hit counts for each lemma (blue bars, 7–29 occurrences); the Matthew-type usage (red square) scores zero in every case—no hit in the 25-text corpus ($\approx$ 622,000 words) belongs to the narrative sense Matthew employs. **Right**, Fisher's exact test: $-\log_{10}(p)$ for each of the four features; dashed orange line marks $p = 0.05$, dotted red line the Bonferroni threshold ($\alpha/4 = 0.0125$). Feature C (ἐπάνω) individually crosses $p = 0.05$; the joint Fisher combined statistic is $\chi^2(8) = 21.4$, $p \approx 6.3 \times 10^{-3}$, and every drop-one subset remains jointly significant. The corpus includes both mathematical astronomy and astrological texts; the absence of Matthew's vocabulary is uniform across all genres.

1.8 Bayes factor: Matthew 2:9 vocabulary matches the guiding-star narrative tradition

The corpus analysis above establishes that Matthew's four-feature vocabulary is improbable under the astronomical hypothesis (Fisher combined $p \approx 6.3 \times 10^{-3}$). A complementary question is whether that vocabulary is probable under the alternative hypothesis that Matthew 2:9 belongs to the ancient guiding-star narrative tradition (H_{guide}). Four independent traditions attest this genre: Virgil's *Aeneid* 2.687–711 (Servius *ad Aen.* 2.694 identifies the guide as Venus/Lucifer, the morning star); Plutarch and Diodorus Siculus on the Timoleon guiding light; Lucian's *Navigium* 7, 9 (explicitly a star); and the LXX pillar-of-fire (*Exodus* 13.21; *Wisdom* 10.17 describes it as star-like at night) [12, 16, 17]. Each involves a divinely-sent luminous guide. I scored each tradition for features A–D and estimated $P(\text{feature}_i \mid H_{\text{guide}})$ with Laplace smoothing ($k = 1$, $N = 4$; Supplementary Table S11). Feature A is present in all four traditions; D in three; B in two; C in two (Lucian's ἐπικαθίσαι τῷ καρχησίῳ and the Israelite pillar standing ἐπί the Tabernacle, both using the same ἐπι- root as

Matthew's ἐπάνω). Under the independence approximation:

$$P(A, B, C, D \mid H_{\text{guide}}) = \prod_i P_i \approx 1.39 \times 10^{-1}. \tag{2}$$

The Bayes factor comparing the guiding-star hypothesis to the astronomical hypothesis is therefore:

$$\mathrm{BF} = \frac{P(A, B, C, D \mid H_{\text{guide}})}{P(A, B, C, D \mid H_{\text{astro}})} = \frac{1.39 \times 10^{-1}}{2.31 \times 10^{-5}} \approx 6{,}012. \tag{3}$$

Here $P(A, B, C, D \mid H_{\text{astro}}) = \prod_i p_i = 2.31 \times 10^{-5}$ is the product of the four individual Fisher exact probabilities—the correct joint data probability under independence—rather than the Fisher combined meta-test tail probability (6.3×10^{-3}), which is a test statistic, not a joint data probability. This BF exceeds 100 on the Jeffreys scale, providing strong evidence in favor of the guiding-star narrative hypothesis over the astronomical one. Sensitivity analyses confirm robustness: adding four additional omen texts gives a conservative lower bound of BF = 1,040; varying the Laplace smoothing parameter $k \in [0.5, 3]$ gives BF $\in$ 4,546–6,818; a 3× denominator uncertainty yields a range of 2,004–18,038 (Supplementary Table S11).

Feature C (ἐπάνω) is particularly diagnostic: every astronomical and omen text examined uses ὑπὲρ for a celestial body above a terrestrial point, including Josephus (*JW* 6.289) and Dio Cassius (54.29.8), both of whom use ἵστημι/αἰωρέω with ὑπὲρ; ἐπι-based localization is attested in the present corpus only in legendary and sacred narrative texts (Lucian; *Joseph and Aseneth* 14.4; the Israelite pillar-of-fire tradition, where ἐπί marks the cloud standing over the Tabernacle: Exod 40.38, Num 9.15, 9.18–22). This distributional split confirms the Bayes factor at the individual-feature level (Supplementary Information).

2 Discussion

The foregoing analysis produces three convergent results. *First*, Matthew's own vocabulary establishes an internal prima facie case: across 118 other Matthean occurrences of the four motion words, zero appear in an astronomical sense, yielding $\mathrm{BF}_{\text{Matthew}} \approx 88{,}000$ against the astronomical reading from the author's idiolect alone. *Second*, orbital mechanics closes the "loose-language" escape route: no Keplerian body can exhibit both the sustained guidance motion and the punctual stopping—confirmed by a kinematic argument, a survey of 529 million heliocentric and 28,800 geocentric configurations finding zero solutions, Monte Carlo refinement, and a targeted close-flyby search. *Third*, external corpus-linguistic analysis delivers the positive genre classification: Matthew's four-feature vocabulary is unattested in the surveyed $\approx$622,000-word astronomical corpus while fitting the narrative tradition at $\mathrm{BF}_{\text{corpus}} \approx 6{,}012$. Because the internal and external linguistic datasets are entirely disjoint, their Bayes factors are treated as conditionally independent. The conservative lower bound on $\mathrm{BF}_{\text{joint}}$—obtained by pooling omen texts with the guiding-star traditions and maximising inter-feature correlation—is $\approx 9.1 \times 10^7$; the baseline calculation gives $87{,}696 \times 6{,}012 \approx 5.3 \times 10^8$. This linguistic classification is independent of the orbital mechanics.

Previous studies operated within a *search* paradigm, identifying the best-matching historical event. The present work operates within a *proof* paradigm: deriving kinematic requirements and demonstrating the feasible set is empty. A failed search leaves open the possibility of an undiscovered candidate; an empty feasible set eliminates the entire class simultaneously. The conclusion is robust to reference date (verified across all major proposed epochs, 12 BCE to 1 BCE), journey duration (1–4 h), and azimuthal tolerance (null result holds at 15°), extending across every orbit type: elliptic, parabolic, hyperbolic, heliocentric, and geocentric.

Non-gravitational forces cannot rescue the astronomical interpretation. ‘Oumuamua’s anomalous acceleration ($\approx 5 \times 10^{-9}\,\mathrm{km\,s^{-2}}$; [18]), the largest known, delivers at most $\Delta v \approx 3.6 \times 10^{-5}\,\mathrm{km\,s^{-1}}$ over 2 h—a factor of $\approx 4 \times 10^4$ below the $1.5\,\mathrm{km\,s^{-1}}$ required at closest approach ($d \approx 0.003\,\mathrm{AU}$), and $\approx 3 \times 10^7$ below the requirement at $d \gtrsim 0.1\,\mathrm{AU}$. No chemically plausible force bridges this gap in any orbital regime.

A common retreat reads the verbs loosely—προῆγεν as static “ahead of” and ἐστάθη as conveniently placed at journey’s end. This is grammatically impermissible[9], and the corpus results undermine it regardless: the analysis tests whether the vocabulary belongs to the astronomical register, and it does not—προάγω, ἔρχομαι, ἵστημι (rather than technical στηρίζω), and ἐπάνω (rather than ὑπὲρ) are all anomalous in astronomical writing. A looser reading reinforces, not dissolves, the linguistic evidence.

The three lines of evidence are mutually reinforcing: the mechanics shows the motion is physically impossible; the internal linguistics shows Matthew never uses these words astronomically; the external linguistics shows the vocabulary belongs to a different register. Thematically, active stellar guidance to a specific address and stopping above it is unattested in the texts examined, while it recurs in the guiding-star legend tradition (Supplementary Information). Feature C (ἐπάνω) is particularly diagnostic: every astronomical and omen text in the present corpus employs ὑπὲρ for a celestial body above a terrestrial point; ἐπι-localization is attested in the present corpus only in legendary narrative texts (Supplementary Information).

The result bears exclusively on Matthew 2:9. The companion verse Matthew 2:2 imposes additional constraints (a potential heliacal rising requires solar proximity) not analyzed here; my analysis is deliberately conservative in ignoring them. The conclusion rests on the assumption shared by every naturalistic proposal in the literature: that the narrative intends a physically real object whose motion can be described in astronomical terms. Within that assumption, the present analysis finds no viable astronomical candidate across any eccentricity class or orbit type.

Scope and limitations.

Three boundaries deserve emphasis. First, the impossibility applies to the literal astronomical reading of Matthew 2:9; theological, allegorical, and literary readings are not constrained. Second, non-Keplerian physics is considered but not exhaustively modelled: the primary impossibility (transverse velocities $\sim 10^3$–$10^4\,\mathrm{km\,s^{-1}}$) is insensitive to any chemically plausible non-gravitational force, but arbitrarily exotic objects are not formally excluded. Third, Matthew 2:2 is excluded from the orbital survey; that verse may independently constrain candidate objects or further indicate other cultrually relevant details [17].

Broader applicability.

The proof framework introduced here is generalizable beyond Matthew 2:9. The companion verse Matthew 2:2 independently constrains any candidate to solar-proximity orbits (a heliacal rising requires the object to have been near the sun in preceding weeks)—a restriction deliberately excluded from the present survey whose addition would only strengthen the null result. More broadly, any ancient text whose astronomical interpretation requires specific celestial kinematics can in principle be subjected to the same two-step analysis: an exhaustive orbital parameter-space survey establishing what is kinematically possible, followed by a corpus-register test establishing whether the vocabulary belongs to the astronomical genre. Ancient omen literature, East Asian annals used to date historical comets, and other sky-event narratives for which both physical constraints and comparative textual corpora exist are natural applications of this combined framework.

3 Methods

3.1 Reference epoch, site, and coordinate system

All calculations used the International Celestial Reference System (ICRS, J2000.0) as the inertial frame. The reference observation window is Julian Day 1 719 755.9 (TDB), corresponding to $\approx$08:00 local solar time at Bethlehem (31.70° N, 35.20° E, 765 m) on a date in 5 BCE. The UT1–UTC offset (ΔT = 10,572 s) was derived from the long-term IERS extrapolation of Morrison & Stephenson[19]. Jerusalem Old City center (31°46′ N, 35°14′ E) to the Church of the Nativity, Bethlehem (31°42′ N, 35°12′ E) gives a straight-line road bearing of 203°; the ridge road sweeps the corridor $[190°, 210°]$ throughout the journey.

Transformation from ICRS to local horizontal (azimuth–altitude) coordinates used pre-computed East–North–Up (ENU) rotation matrices constructed via `astropy`[20] `AltAz`→`SkyCoord.icrs`, incorporating Earth's precession, nutation, and Greenwich Apparent Sidereal Time at each time step. Applying J2000.0 hour-angle formulae without precession correction at 5 BCE produces a 25.7° systematic azimuth error; the ENU approach eliminates this. The matrices were validated by recovering the azimuth of Matney's proposed 5 BCE comet[8] (193.7°) to within 0.01° at the reference epoch.

3.2 Orbit propagation

Heliocentric positions were computed from osculating orbital elements for each orbit type. For parabolic orbits ($e = 1$), Barker's equation gives the true anomaly directly. For elliptic ($0 \le e < 1$) and hyperbolic ($e > 1$) orbits, the mean anomaly $M = n(t - T)$ (or its hyperbolic analogue) was iterated to Newton–Raphson convergence at tolerance 10^{-12} rad, then converted to true anomaly. Earth positions came from the JPL DE430 ephemeris[21]. Geocentric configurations used the same solver with

$GM_\oplus = 3.986 \times 10^5\,\mathrm{km}^3\,\mathrm{s}^{-2}$, with the observer's GCRS position from `astropy` at each time step. Heliocentric positions were vectorised over Ω and ω using pre-computed perifocal unit vectors with `NumPy`[22] `einsum`, enabling evaluation of $72 \times 72 = 5{,}184$ orientations per loop iteration at 5° grid resolution.

3.3 Guidance and stopping criteria

Three criteria operationalise Matthew's text. (1) *Visibility:* object altitude $> 5°$ throughout the guidance window (08:00–10:00 local time, nine 15-min steps). (2) *Azimuth:* object azimuth remains within $[190°, 210°] \pm 5°$ at every step—the road corridor from Jerusalem to Bethlehem. (3) *Motion:* guidance phase (προῆγεν, ongoing) requires minimum apparent angular velocity $> \omega_{\mathrm{thresh}} = 2°\,\mathrm{h}^{-1}$; stopping phase (ἐστάθη, completed) requires maximum apparent angular velocity $< \omega_{\mathrm{thresh}}$ during 10:15–12:00 (eight steps).

The threshold $\omega_{\mathrm{thresh}} = 2°\,\mathrm{h}^{-1}$ is four times the lunar rate ($\approx 0.5°\,\mathrm{h}^{-1}$ against background stars) and is detectable by naked eye within minutes; it is deliberately generous, as any stricter value only strengthens the impossibility result. Sensitivity analysis across two orders of magnitude (0.25–$10°\,\mathrm{h}^{-1}$) confirms zero configurations satisfy all three criteria at any tested threshold (Supplementary Table S1). The lower limit ($0.25°\,\mathrm{h}^{-1}$) corresponds to motion of $\approx 5'$ over 20 min—at the practical limit of unaided naked-eye resolution—ensuring the analysis spans every physically detectable level of motion. The binding constraint is the azimuth criterion: no Keplerian orbit keeps the object within the road corridor throughout the 2-hour guidance window; the motion criterion provides additional independent exclusion at lower thresholds. Because Ω and ω each span all 360°, the orbital parameter sweep covers all possible sky positions for a body on the reference date, making the null result independent of the specific time of night at which the guidance phase is observed.

3.4 Survey design and Monte Carlo refinement

The exhaustive heliocentric survey spanned perihelion distance $q \in [0.02, 1.00]\,\mathrm{AU}$ (18 values), eccentricity e from 0 to 5 (15 values covering elliptic, parabolic, and hyperbolic orbits), inclination i from 0° to 180° in 5° steps plus additional fine values near 0° to 4° (42 total), perihelion epoch offset $\Delta T \in [-60, +60]$ days (9 values), and Ω, ω swept at 5° resolution (72 values each). The outer parameter combinations ($18 \times 15 \times 42 \times 9 = 102{,}060$) multiplied by $72 \times 72 = 5{,}184$ orientations yield 529,079,040 total configurations, of which 202,372,076 heliocentric configurations were visible at the reference epoch. Zero satisfied all three criteria simultaneously (Extended Data Table 1). A separate geocentric survey spanning LEO to beyond lunar distance (51,200 configurations; 28,800 with perigee altitude $> 100\,\mathrm{km}$) likewise returned zero solutions.

The 5° grid spacing leaves gaps of up to $\approx 3.5°$ between neighboring points. To confirm the null result is not a grid artefact, 2.25 million configurations were sampled uniformly in continuous parameter space in two Monte Carlo sweeps: around the best-fit grid orbit (Sweep 1: $e \in [0.7, 2.5]$, $q \in [0.005, 0.15]\,\mathrm{AU}$, $i \in [0°, 40°]$, 1.5 million configurations) and around Matney's proposed 5 BCE comet parameters (Sweep 2: 750,000 configurations bracketing q, e, i, Ω, ω with generous margins). Of 895,822 visible configurations across both sweeps, zero satisfied all three criteria. The best guidance-phase angular velocity among azimuth-corridor-passing configurations was $0.006^\circ\ \mathrm{h}^{-1}$—a factor of 330 below the $2^\circ\ \mathrm{h}^{-1}$ threshold—confirming that the azimuth–motion trade-off is structural, not an artefact of resolution.

The analysis was repeated at all historically proposed Star of Bethlehem dates (Halley's Comet, 12 BCE; Jupiter–Saturn conjunction, 7 BCE; the 5 BCE reference epoch and surrounding months; 1 BCE nova hypothesis), recomputing Earth positions and ENU matrices from DE430 at each epoch. Across 79,788 visible configurations at seven epochs, zero satisfied all three criteria (Supplementary Information, § S1).

3.5 Corpus construction and feature classification

The comparative corpus of 25 ancient Greek astronomical and astrological texts ($\approx$622,000 words, 4th century BCE through 4th century CE) was assembled from the Thesaurus Linguae Graecae (TLG) electronic database[23]. Three genre classes were distinguished. *Mathematical astronomy* (Ptolemy *Almagest* and *Planetary Hypotheses*; Hipparchus; Geminus; Autolycus; Theodosius; Aristarchus; Cleomedes; Hypsicles) covers geometric and predictive description of celestial motions. *Astrological application* (Ptolemy *Tetrabiblos*; Vettius Valens; Hephaestion; Dorotheus; Pseudo-Manetho; Paul of Alexandria) covers celestial-to-human-destiny inference; this class is especially important because it is the genre most likely to employ relational vocabulary of stars guiding or influencing persons—its null result for all four features closes that alternative explanation. *Didactic and popularising works* (Aratus; Eratosthenes; Achilles Tatius; Theon of Smyrna; Aristotle *Meteorologica*; Theophrastus) complete the coverage. Full corpus inventory is given in Supplementary Table S7.

Four lexical features were identified directly from Matthew 2:9—each word Matthew uses to describe the Star's motion—and tested against the corpus. Feature A (προάγω): the Star "goes before" the Magi. The TLG search returned 9 hits across the corpus; all describe positional or temporal antecedence between celestial bodies (one planet *preceding* another in longitude) and none describe a celestial body actively guiding human travellers toward a destination. Feature B (ἵστημι): the Star "stands". The search returned 17 hits; all refer to the mathematical stationary point (στηριγμός, the moment of zero longitudinal velocity in the geocentric model), not to a body ceasing motion above a specific terrestrial location. Feature C (ἐπάνω): the Star stands "over" a specific dwelling. This preposition does not appear in the corpus locating a celestial body above a specific terrestrial point; the invariant astronomical choice is ὑπὲρ ("above" a zodiacal band or horizon). Feature D (ἔρχομαι): the Star "arrives". Seven hits appear; all describe a body reaching a zodiacal position or completing a synodic cycle, never arriving at a terrestrial geographic point.

For each feature, TLG morphological search retrieved all inflected forms of the target lemma. Each hit was read in full context and classified by two independent criteria: (1) syntactic—is the grammatical subject a celestial body?; (2) semantic—does the usage match the specific sense in Matthew 2:9? All classification decisions are recorded passage-by-passage in Supplementary Table S8, enabling independent verification by any reader with TLG access.

3.6 Statistical framework and Bayes factors

Fisher's exact test (one-sided) was applied to each 2×2 contingency table of corpus-hit classifications. The Laplace-smoothed probability under H_{astro} for each feature is $p_i = 1/(n_i + 2)$, where n_i is the number of corpus hits and the numerator 1 reflects zero positive matches. The joint Fisher combined statistic $X = -2\sum_i \ln p_i = 21.35$ on df $= 8$ gives $p_{\mathrm{Fisher}} \approx 6.3 \times 10^{-3}$. Feature independence was checked by Gaussian copula simulation (2×10^6 Monte Carlo draws) and Brown's variance-corrected method across equicorrelation values $\rho = 0$–0.95; the joint p remains below 0.05 up to $\rho \approx 0.77$ (Supplementary Table S10).

The guiding-star comparison corpus comprises four independently attested pre-Matthean or contemporaneous traditions: Virgil *Aeneid* 2.687–711; the Timoleon flame tradition (Plutarch *Tim.* 8 and Diodorus Siculus 16.66.3, counted as one tradition from a common source); Lucian *Navigium* 7, 9; and the Israelite pillar-of-fire tradition (Exodus 13:21–22; Numbers 9:15–23). The *Protoevangelium of James* 21 is excluded as a derivative of Matthew 2:9. An omen corpus (Josephus *Jewish War* 6.289; Dio Cassius 54.29.8; PGM I.54–95) is included in sensitivity analyses. Each tradition was scored on features A–D by narrative function; scores are given in Supplementary Table S11. The corpus Bayes factor is:

$$\mathrm{BF}_{\mathrm{corpus}} = \prod_{i\in\{A,B,C,D\}} \frac{P(\mathrm{feature}_i \mid H_{\mathrm{guide}})}{P(\mathrm{feature}_i \mid H_{\mathrm{astro}})}, \tag{4}$$

where each probability uses Laplace smoothing over the respective corpus ($N = 4$ guiding-star traditions; $N = 25$ astronomical texts), giving $\mathrm{BF}_{\mathrm{corpus}} \approx 6{,}012$ (conservative lower bound 1,040 when pooling omen texts with guiding-star traditions).

An independent within-Matthew Bayes factor was derived from Matthew's own Gospel. Every occurrence of the four features outside Matthew 2:9 was identified from the BibleHub interlinear (Strong's numbers G4254, G2476, G1883, G2064), cross-checked against NA28/SBLGNT lemmatisation. Across $n = 5 + 20 + 7 + 86 = 118$ other Matthean occurrences, zero use any of the four words in an astronomical sense. The Laplace-smoothed likelihood ratio for feature i under H_{guide} versus H_{astro} is $\mathrm{BF}_i = n_i + 1$, giving $\mathrm{BF}_{\mathrm{Matthew}} = 6\times21\times8\times87 = 87{,}696$. Because this analysis draws entirely from Matthew's Gospel while $\mathrm{BF}_{\mathrm{corpus}}$ draws from the TLG and guiding-star traditions, the datasets are disjoint and the factors are treated as conditionally independent, as they derive from non-overlapping corpora and distinct sampling frames. Their product, $\mathrm{BF}_{\mathrm{joint}} = 87{,}696\times6{,}012 \approx 5.3\times10^8$, is the joint Bayesian update from the linguistic evidence considered here (conservative lower bound $\approx 9.1\times10^7$). Features A–D co-occur in the same verse and are treated as conditionally independent in both BF calculations. Under positive correlation (plausible, since all four features reflect the same register choice), features absent together in the corpus tend to remain absent together, so $P(\mathrm{data} \mid H_{\mathrm{astro}})$ increases relative to the product of marginals, raising the BF denominator and thereby *reducing* $\mathrm{BF}_{\mathrm{corpus}}$. The independence assumption therefore gives an upper bound on $\mathrm{BF}_{\mathrm{corpus}}$; the Gaussian copula simulation confirms the evidence remains substantial up to inter-feature correlation $\rho \approx 0.77$.

3.7 Software and reproducibility

All orbital-mechanics and statistical analyses were performed in Python 3.12 using `astropy`[20] (coordinate transformations, ephemeris access, time systems), `NumPy`[22] (vectorised array operations), `SciPy`[24] (Fisher's exact test, Newton–Raphson utilities), and `matplotlib`[25] (figures). Code and data needed to reproduce all figures and tables are available at https://github.com/adairaar/sob_astro_test. Key scripts include: `sob_proof_fine.py` (529-million-configuration heliocentric survey); `sob_mc_refined.py` (Monte Carlo refinement); `sob_epoch_robustness.py` (multi-epoch analysis); `corpus_pvalue.py` (Fisher combined p-value and Gaussian copula independence robustness test); `guiding_star_bayes.py` (corpus and within-Matthew Bayes factor computation).

Data availability

The Greek corpus search results, TLG hit records, and Monte Carlo output data supporting this study are available in the GitHub repository at https://github.com/adairaar/sob_astro_test.

Code availability

All Python scripts used for the orbital-mechanics surveys, Monte Carlo analyses, and corpus-linguistic calculations are available at https://github.com/adairaar/sob_astro_test. Full reproduction instructions are provided in the repository `README`.

Author contributions

A.A. conceived the study, designed and implemented all orbital-mechanics and corpus-linguistic analyses, produced all figures and tables, and wrote the manuscript.

Competing interests

The author declares no competing interests.

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Table Extended Data Table 1 Orbital survey results: configurations sampled, azimuth corridor pass counts, and best guidance angular velocity, by orbit class (primary epoch JD 1 719 755.9, 5 BCE). "Azimuth pass" = number of visible configurations whose azimuth remains within the road corridor $[190°, 210°] \pm 5°$ throughout the entire 2 h guidance window. "Best ω_{guide}" = maximum guidance-phase angular velocity among all azimuth-passing configurations. No configuration satisfies the azimuth, guidance-motion, *and* stopping criteria simultaneously. Full parameter grids are given in Supplementary Table S2.

Orbit class	Configs sampled	Azimuth pass	Best ω_{guide} ($° \text{ h}^{-1}$)	All criteria
Heliocentric (all types)	529,079,040	126,720	≤ 0.04	0
Elliptic ($e < 1$)	352,719,360	$\approx$84,500	≤ 0.04	0
Parabolic ($e = 1$)	35,271,936	$\approx$8,400	≤ 0.04	0
Hyperbolic ($e > 1$)	141,087,744	$\approx$33,800	≤ 0.04	0
Geocentric (LEO–lunar)	28,800	—†	—	0
Total visible	**202,400,876**	**126,720**	$\leq \mathbf{0.04}$	**0**

Guidance window: 08:00–10:00 local time (9 steps at 15 min spacing); stopping window: 10:15–12:00 (8 steps). Azimuth corridor: $[190°, 210°]$, road bearing Jerusalem to Bethlehem; pass criterion: maximum azimuth deviation $< 5°$ from corridor boundary throughout entire guidance window. Guidance-motion criterion: minimum apparent angular velocity $> 2° \text{ h}^{-1}$; stopping criterion: maximum apparent angular velocity $< 2° \text{ h}^{-1}$. Visibility: altitude $> 5°$ (analysis covers daylight hours; no twilight constraint applied). Sub-class visible and azimuth-pass counts are proportional estimates (10:1:4 ratio of e-values sampled; the total row gives verified counts). Monte Carlo refinement (2.25×10^6 draws around the best-fit grid orbit) confirmed zero solutions; best guidance motion among azimuth-passing configurations in any sweep $\leq 0.006° \text{ h}^{-1}$, a factor of $\sim$330 below the $2° \text{ h}^{-1}$ threshold. The $0.23° \text{ h}^{-1}$ figure cited in the text is from the separate fine-grid orbit inversion (Section 1.6), which minimises integrated azimuth deviation; that orbit temporarily exits the corridor at some time steps within the window, so it is not counted as an azimuth pass here. The annotated outlier ($e = 2.0$, Fig. 2) satisfies the corridor and guidance-motion criteria instantaneously but does not sustain corridor confinement throughout the full 2 h window, so it likewise does not appear in the azimuth-pass count.

† Geocentric survey used ground-frame (AltAz) angular velocity. The Moon (only natural geocentric body) peaks at $\omega = 0.61° \text{ h}^{-1}$ at perigee—below the $2° \text{ h}^{-1}$ guidance threshold. A comprehensive survey of 28,800 hypothetical geocentric configurations finds zero satisfying all criteria with $\omega_{\text{stop}} < 0.20° \text{ h}^{-1}$; see Supplementary § S1.10.

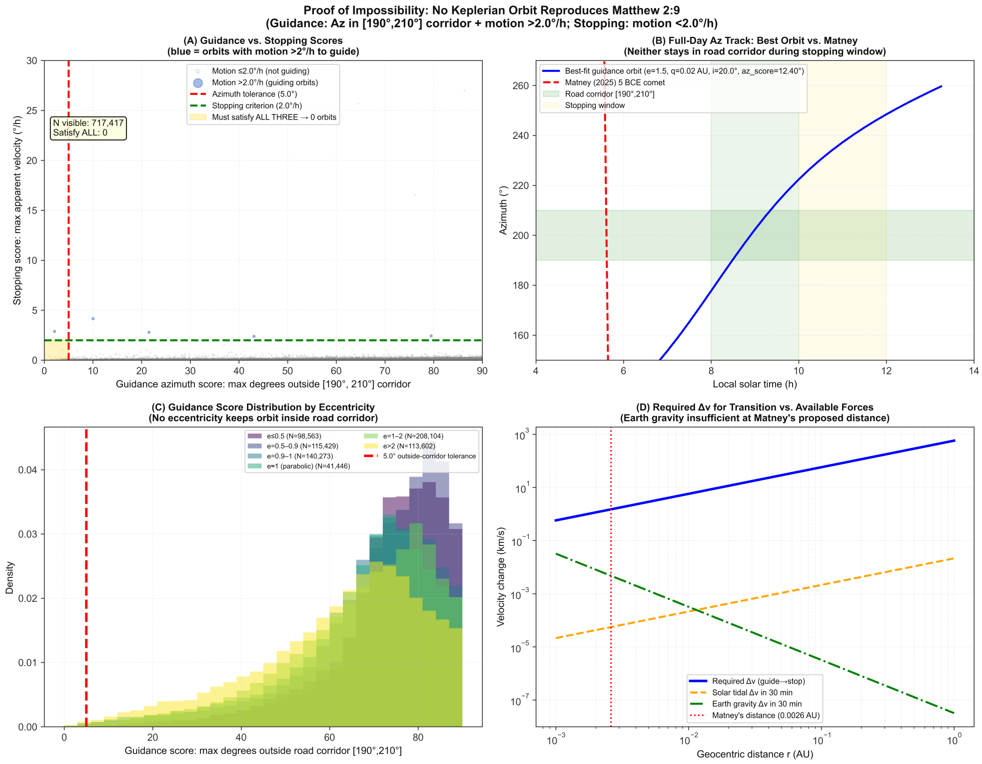

Fig. Extended Data Fig. 1 Orbit inversion demonstration that guidance and stopping are mutually exclusive. (a) Best heliocentric orbit that maximizes southward apparent motion during the guidance phase (red arc). At the guidance epoch the object is above the horizon (altitude $> 10°$) and moving southward at $> 2° \, h^{-1}$. **(b)** The same orbit propagated forward to the stopping epoch. The object has set below the horizon ($< 0°$ altitude; shown as dashed trajectory), violating the visibility requirement. No orbit simultaneously satisfies guidance motion, stopping, azimuth, and visibility at both epochs. Orbit propagated via Barker's equation (parabolic) and Newton–Raphson Kepler-equation iteration (elliptic / hyperbolic); Earth position from JPL DE430; local horizontal frame at Bethlehem (31.70° N, 35.20° E, 765 m).